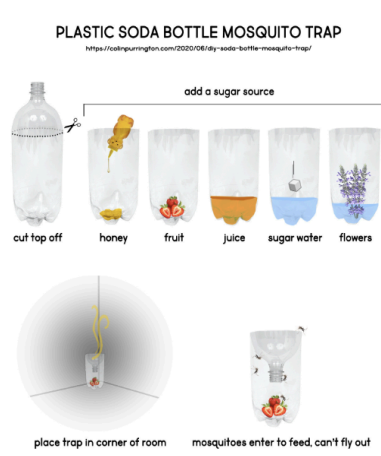


Element B: Documentation and Analysis of Prior Solution Attempts

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

To address the insect infestation in Student E's bedroom, it is important to examine prior solutions that commonly manage these problems. The first solution uses a cut plastic bottle filled with a sweet liquid attractant, such as juice or honey, to lure and trap flying insects like mosquitoes and small flies (*How to Make a DIY Wasp Trap That Actually Works*, 2025). This solution is simple and inexpensive to make, using materials that are commonly available in most households. It is effective for smaller insects and provides an eco-friendly alternative to chemical-based traps. However, this method does not work well for larger insects. It can also take time to show results and requires regular cleaning and bait refills to maintain effectiveness. Additionally, it may attract more pests than it eliminates if not closely monitored.



Indoor insect trap made out of a cut water bottle

Another common solution involves using an inverted plastic bottle with a wasp or hornet attractant. This design traps the insect inside and prevents escape, often near outdoor or indoor

entry points. It is easy to make using common household materials and offers a chemical-free solution for pest control. It is especially effective at reducing the presence of stinging insects in specific areas. However, it may also capture helpful or non-target insects. Like the previous solution, it must be cleaned and have its bait changed. There is also a risk of drawing in more pests before the trap begins to work, which makes the problem worse if not properly managed (*DIY Wasp Trap • Everyday Cheapskate, 2023*).



DIY wasp trap

A third common solution is peppermint oil, a natural insect repellent. Peppermint oil is safe for humans, works against many insects, and has a pleasant scent. It is easy to use and can serve multiple household purposes. While this method is attractive due to its safety and simplicity, it is not a permanent solution. It requires frequent re-application to remain effective and can cause allergic reactions in some individuals. In large quantities, peppermint oil is toxic to pets such as cats and dogs, making it a less suitable long-term solution in shared living spaces (*Non-Toxic Insect Control: The Benefits of Peppermint Oil, 2025*).



Peppermint oil

While each of these prior attempts offers strengths, they also present drawbacks that prevent them from fully meeting the needs of Student E.

Currently, no active solutions are being used to manage the insect problem in Student E's bedroom. While prior attempts, like peppermint oil and homemade bottle traps, were used in the past for individuals facing the same problem, they were found to be ineffective.



References

Purrington, Colin. "Indoor Mosquito Trap Made from Plastic Soda Bottle » Colin Purrington's

Blog." Colin Purrington's Blog, 22 June 2020,

<https://colinpurrington.com/2020/06/diy-soda-bottle-mosquito-trap/>.

"How to Make a DIY Wasp Trap That Actually Works." The Spruce,

<https://www.thespruce.com/how-to-make-a-wasp-trap-1389067>. Accessed 29 Apr. 2025.

Peppermint oil: Uses, benefits, side effects, and safety risks. (2019, April 25). Healthline.

<https://www.healthline.com/health/benefits-of-peppermint>

Hunt, Mary. "DIY Wasp Trap • Everyday Cheapskate." Everyday Cheapskate, 9 June 2023,

<https://www.everydaycheapskate.com/diy-wasp-trap-that-really-works-and-other-great-re-ader-tips/>.

"Non-Toxic Insect Control: The Benefits of Peppermint Oil." Dr. Killigan's,

<https://drkilligans.com/blogs/ingredients/non-toxic-insect-control-the-benefits-of-peppermint-oil>. Accessed 6 May 2025.